

Mathematical Foundations of the WCS-GNCCP Molecular Graph Kernel

I. PROBLEM FORMULATION

Let G and H be two molecular graphs with M and N atoms, respectively. Their topology is encoded by weighted adjacency matrices $\mathbf{A}_G \in \mathbb{R}^{M \times M}$ and $\mathbf{A}_H \in \mathbb{R}^{N \times N}$, where each entry reflects bond type (single, double, triple, aromatic) and topology (ring membership, conjugation). A cost matrix $\mathbf{C} \in \mathbb{R}^{M \times N}$ encodes pairwise atom-feature dissimilarity.

Definition 1 (Partial Permutation Polytope). *For an integer $L \leq \min(M, N)$, the partial permutation polytope is*

$$\mathcal{P}_L = \{\mathbf{X} \in \{0, 1\}^{M \times N} \mid \mathbf{X}\mathbf{1}_N \leq \mathbf{1}_M, \mathbf{X}^T \mathbf{1}_M \leq \mathbf{1}_N, \mathbf{1}_M^T \mathbf{X}\mathbf{1}_N = L\}, \quad (1)$$

and its convex relaxation $\mathcal{D}_L$ replaces $\{0, 1\}^{M \times N}$ with $[0, 1]^{M \times N}$. An entry $X_{ij} = 1$ indicates that atom i in G is matched to atom j in H .

Definition 2 (WCS Objective). *The Weighted Common Sub-graph (WCS) objective is*

$$F(\mathbf{X}) = \alpha f(\mathbf{X}) + (1 - \alpha) \text{tr}(\mathbf{C}^T \mathbf{X}), \quad (2)$$

where $\alpha \in [0, 1]$ balances structure against atom features,

$$f(\mathbf{X}) = \|\mathbf{U} \circ \mathbf{A}_G - \mathbf{X}\mathbf{A}_H\mathbf{X}^T\|_F^2 \quad (3)$$

is the structural mismatch term, $\circ$ denotes the Hadamard product, and

$$\mathbf{U} = \mathbf{X}\mathbf{1}_{N \times N}\mathbf{X}^T \in \mathbb{R}^{M \times M} \quad (4)$$

is the co-matching indicator.

II. COMPUTATIONAL IDENTITY FOR $\mathbf{U}$

The matrix $\mathbf{U}$ in (4) involves the $N \times N$ all-ones matrix, requiring $O(MN^2 + M^2N)$ work. We show that $\mathbf{U}$ admits a rank-one factorization that reduces this to $O(MN + M^2)$.

Lemma 1 (Rank-One Factorization of $\mathbf{U}$). *Let $\mathbf{r} = \mathbf{X}\mathbf{1}_N \in \mathbb{R}^M$ be the row-sum vector. Then*

$$\mathbf{U} = \mathbf{X}\mathbf{1}_{N \times N}\mathbf{X}^T = \mathbf{r}\mathbf{r}^T. \quad (5)$$

Proof. Factor the all-ones matrix as $\mathbf{1}_{N \times N} = \mathbf{1}_N \mathbf{1}_N^T$:

$$\begin{aligned} \mathbf{U} &= \mathbf{X}(\mathbf{1}_N \mathbf{1}_N^T)\mathbf{X}^T \\ &= (\mathbf{X}\mathbf{1}_N)(\mathbf{X}\mathbf{1}_N)^T = \mathbf{r}\mathbf{r}^T. \end{aligned} \quad (6)$$

□

Remark 1. $\mathbf{U}$ is symmetric PSD with rank one. For binary $\mathbf{X} \in \mathcal{P}_L$ each $r_i \in \{0, 1\}$, so $U_{ik} = 1$ iff both atoms i, k are matched.

III. GRADIENT OF THE STRUCTURAL TERM

We derive $\nabla_{\mathbf{X}} f$ for the structural cost (3).

Theorem 1 (Structural Gradient). *Define the residual $\mathbf{R} = \mathbf{U} \circ \mathbf{A}_G - \mathbf{X}\mathbf{A}_H\mathbf{X}^T$. Then*

$$\nabla_{\mathbf{X}} f = 4(\mathbf{R} \circ \mathbf{A}_G)\mathbf{X}\mathbf{1}_{N \times N} - 4\mathbf{R}\mathbf{X}\mathbf{A}_H. \quad (7)$$

Moreover, using Lemma 1:

$$(\mathbf{R} \circ \mathbf{A}_G)\mathbf{X}\mathbf{1}_{N \times N} = [(\mathbf{R} \circ \mathbf{A}_G)\mathbf{X}]\mathbf{1}_N \mathbf{1}_N^T, \quad (8)$$

a rank-one matrix whose i -th row repeats $\sum_j [(\mathbf{R} \circ \mathbf{A}_G)\mathbf{X}]_{ij}$.

Proof. Write $f = \text{tr}(\mathbf{R}^T \mathbf{R})$. By the chain rule,

$$df = 2 \text{tr}(\mathbf{R}^T d\mathbf{R}). \quad (9)$$

Part 1 (Hadamard term). Since $\mathbf{A}_G$ is constant,

$$d(\mathbf{U} \circ \mathbf{A}_G) = (d\mathbf{U}) \circ \mathbf{A}_G. \quad (10)$$

From $\mathbf{U} = \mathbf{r}\mathbf{r}^T$ with $\mathbf{r} = \mathbf{X}\mathbf{1}_N$:

$$d\mathbf{U} = (d\mathbf{X}\mathbf{1}_N)\mathbf{r}^T + \mathbf{r}(d\mathbf{X}\mathbf{1}_N)^T. \quad (11)$$

Part 2 (Bilinear term). By the product rule,

$$\begin{aligned} d(\mathbf{X}\mathbf{A}_H\mathbf{X}^T) &= (d\mathbf{X})\mathbf{A}_H\mathbf{X}^T \\ &\quad + \mathbf{X}\mathbf{A}_H(d\mathbf{X})^T. \end{aligned} \quad (12)$$

Part 3 (Combining). Substitute (10)–(12) into (9) and apply the Hadamard trace identity

$$\text{tr}(\mathbf{A}^T(\mathbf{B} \circ \mathbf{C})) = \text{tr}((\mathbf{A} \circ \mathbf{B})^T \mathbf{C}).$$

Collecting terms in $d\mathbf{X}$: the Hadamard contribution yields $(\mathbf{R} \circ \mathbf{A}_G)\mathbf{X}\mathbf{1}_{N \times N}$ and the bilinear contribution yields $-\mathbf{R}\mathbf{X}\mathbf{A}_H$. Since $\mathbf{R}$, $\mathbf{A}_G$, $\mathbf{U}$, and $\mathbf{X}\mathbf{A}_H\mathbf{X}^T$ are all symmetric, each of the two terms in (11) and (12) contributes equally, producing the factor of 4. □

Remark 2. A common error is to differentiate $\mathbf{U} \circ \mathbf{A}_G$ as if it were the matrix product $\mathbf{U}\mathbf{A}_G$. The Hadamard derivative is $d(\mathbf{U} \circ \mathbf{A}_G) = (d\mathbf{U}) \circ \mathbf{A}_G$ (element-wise), not $(d\mathbf{U})\mathbf{A}_G$.

IV. THE GNCCP CONTINUATION METHOD

Direct minimization of F over $\mathcal{D}_L$ is nonconvex. GNCCP introduces $\zeta \in [-1, 1]$ to deform the problem from convex to concave.

Definition 3 (GNCCP Objective). *For $\zeta \in [-1, 1]$, define*

$$J_\zeta(\mathbf{X}) = \begin{cases} (1 - \zeta) F(\mathbf{X}) + \zeta \|\mathbf{X}\|_F^2 & \zeta \geq 0, \\ (1 + \zeta) F(\mathbf{X}) + \zeta \|\mathbf{X}\|_F^2 & \zeta < 0. \end{cases} \quad (13)$$

Proposition 1 (Endpoint Properties).

- 1) $\zeta = +1$: $J_{+1} = \|\mathbf{X}\|_F^2$ is strictly convex. Its unique minimizer over $\mathcal{D}_L$ is the uniform matrix $\mathbf{X}^{(0)} = \frac{L}{MN} \mathbf{1}_{M \times N}$.
- 2) $\zeta = 0$: $J_0 = F(\mathbf{X})$, the original WCS objective.
- 3) $\zeta = -1$: $J_{-1} = -\|\mathbf{X}\|_F^2$ is strictly concave. Its minimizers over $\mathcal{D}_L$ are vertices, i.e., partial permutation matrices in $\mathcal{P}_L$.

Proof. (1) $\|\mathbf{X}\|_F^2 = \sum_{ij} X_{ij}^2$ is strictly convex. The unique minimizer distributes mass uniformly: $X_{ij} = L/(MN)$ satisfies all constraints and minimizes the sum of squares by Schur convexity.

(2) Direct substitution.

(3) $-\|\mathbf{X}\|_F^2$ is strictly concave. A strictly concave function attains its minimum over a compact convex polytope at a vertex. The vertices of $\mathcal{D}_L$ are exactly the elements of $\mathcal{P}_L$. $\square$

Proposition 2 (Gradient of J_ζ).

$$\nabla_{\mathbf{X}} J_\zeta = \begin{cases} (1 - \zeta) \nabla_{\mathbf{X}} F + 2\zeta \mathbf{X} & \zeta \geq 0, \\ (1 + \zeta) \nabla_{\mathbf{X}} F + 2\zeta \mathbf{X} & \zeta < 0, \end{cases} \quad (14)$$

where $\nabla_{\mathbf{X}} F = \alpha \nabla_{\mathbf{X}} f + (1 - \alpha) \mathbf{C}$ and $\nabla_{\mathbf{X}} f$ is given by Theorem 1.

Proof. $\nabla \|\mathbf{X}\|_F^2 = 2\mathbf{X}$. Apply linearity to (13). $\square$

V. FRANK-WOLFE OPTIMIZATION

For each fixed ζ we solve $\min_{\mathbf{X} \in \mathcal{D}_L} J_\zeta(\mathbf{X})$ via Frank-Wolfe (conditional gradient).

Definition 4 (Linear Minimization Oracle). *At iterate $\mathbf{X}_t$, the FW direction is*

$$\mathbf{Y}_t = \arg \min_{\mathbf{Y} \in \mathcal{D}_L} \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{Y} \rangle. \quad (15)$$

Since the minimum of a linear function over a polytope is attained at a vertex, $\mathbf{Y}_t \in \mathcal{P}_L$. The optimal vertex is found by the Hungarian algorithm in $O(\min(M, N)^3)$ time.

Theorem 2 (Frank-Wolfe Duality Gap). *Let J_ζ be convex on $\mathcal{D}_L$ and let $\mathbf{X}^* = \arg \min_{\mathcal{D}_L} J_\zeta$. The Frank-Wolfe gap*

$$g_t = \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}_t - \mathbf{Y}_t \rangle \quad (16)$$

satisfies: (i) $g_t \geq 0$; (ii) $g_t \geq J_\zeta(\mathbf{X}_t) - J_\zeta(\mathbf{X}^)$; (iii) $g_t = 0 \Leftrightarrow \mathbf{X}_t$ is optimal.*

Proof. (i) $\mathbf{Y}_t$ minimizes the linear form over $\mathcal{D}_L$ and $\mathbf{X}_t \in \mathcal{D}_L$, so $\langle \nabla J_\zeta, \mathbf{Y}_t \rangle \leq \langle \nabla J_\zeta, \mathbf{X}_t \rangle$, giving $g_t \geq 0$.

(ii) By convexity:

$$J_\zeta(\mathbf{X}^*) \geq J_\zeta(\mathbf{X}_t) + \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}^* - \mathbf{X}_t \rangle. \quad (17)$$

Since $\mathbf{Y}_t$ is the minimizer and $\mathbf{X}^* \in \mathcal{D}_L$:

$$\langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{X}^* - \mathbf{X}_t \rangle \geq \langle \nabla J_\zeta(\mathbf{X}_t), \mathbf{Y}_t - \mathbf{X}_t \rangle = -g_t. \quad (18)$$

Combining (17)–(18): $g_t \geq J_\zeta(\mathbf{X}_t) - J_\zeta(\mathbf{X}^*)$.

(iii) If $\mathbf{X}_t$ is optimal, first-order optimality gives $\langle \nabla J_\zeta, \mathbf{Y} - \mathbf{X}_t \rangle \geq 0 \forall \mathbf{Y} \in \mathcal{D}_L$, so $g_t = 0$. Conversely, $g_t = 0$ with (ii) gives $J_\zeta(\mathbf{X}_t) \leq J_\zeta(\mathbf{X}^*)$. $\square$

Remark 3. When J_ζ is nonconvex ($\zeta < 0$), the gap (16) is no longer a certified suboptimality bound, but $g_t = 0$ still implies $\mathbf{X}_t$ is a stationary point. We use $g_t < \varepsilon$ as early termination for all ζ .

VI. KERNEL PROPERTIES

Proposition 3 (Graph Similarity Measure). *The exponential mapping*

$$k(G, H) = \exp\left(-\frac{\gamma F_{GH}^*}{L}\right) \quad (19)$$

serves as a powerful similarity measure for topological graph matching. While functions based on optimal-assignment and graph-edit distances are generally not unconditionally positive semi-definite (PSD) in the strict Mercer sense, they represent highly discriminative indefinite kernels. Because algorithms such as k-Nearest Neighbors (k-NN) rely purely on distance rankings rather than inner-product spaces, they sidestep the theoretical convexity constraints of PSD kernels entirely, allowing us to leverage the exact topological discriminative power of F_{GH}^ without eigenvalue manipulation.*

Remark 4. γ controls bandwidth: small γ spreads kernel values (all pairs appear similar), large γ concentrates mass on near-identical pairs. We default to $\gamma = 0.5$ and normalize by $\max(L, 1)$ for size invariance.

VII. SCALE INVARIANCE

Proposition 4 (Frobenius Normalization). *Let $\hat{\mathbf{A}}_G = \mathbf{A}_G / \|\mathbf{A}_G\|_F$ and $\hat{\mathbf{A}}_H = \mathbf{A}_H / \|\mathbf{A}_H\|_F$. Then*

$$\hat{f}(\mathbf{X}) = \|\mathbf{U} \circ \hat{\mathbf{A}}_G - \mathbf{X} \hat{\mathbf{A}}_H \mathbf{X}^T\|_F^2 \leq 2 \quad (20)$$

for all $\mathbf{X} \in \mathcal{D}_L$, independently of the number of bonds.

Proof. By the identity $\|\mathbf{A} - \mathbf{B}\|_F^2 = \|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2 - 2\langle \mathbf{A}, \mathbf{B} \rangle$, it suffices to bound each squared norm.

For the Hadamard term, using $\|\mathbf{A} \circ \mathbf{B}\|_F \leq \|\mathbf{A}\|_{\max} \|\mathbf{B}\|_F$ with $U_{ik} = r_i r_k \leq 1$ for $\mathbf{X} \in \mathcal{D}_L$:

$$\|\mathbf{U} \circ \hat{\mathbf{A}}_G\|_F \leq 1 \cdot \|\hat{\mathbf{A}}_G\|_F = 1. \quad (21)$$

For the bilinear term, since $\|\mathbf{X}\|_2 \leq 1$ for sub-stochastic $\mathbf{X}$:

$$\|\mathbf{X} \hat{\mathbf{A}}_H \mathbf{X}^T\|_F \leq \|\mathbf{X}\|_2^2 \|\hat{\mathbf{A}}_H\|_F \leq 1. \quad (22)$$

Therefore $\hat{f} \leq \|\mathbf{U} \circ \hat{\mathbf{A}}_G\|_F^2 + \|\mathbf{X} \hat{\mathbf{A}}_H \mathbf{X}^T\|_F^2 \leq 2$. $\square$

Remark 5. Without normalization, $f(\mathbf{X})$ scales with $\|\mathbf{A}_G\|_F^2 + \|\mathbf{A}_H\|_F^2$, which grows linearly in the number of

bonds. For 30 bonds at weight ~ 1.5 , $\|\mathbf{A}_G\|_F \approx 9.5$, making the structural cost $\sim 90\times$ larger than the appearance cost $\text{tr}(\mathbf{C}^T \mathbf{X}) \in [0, L]$. Normalization restores the balance intended by α .

VIII. PROJECTION ONTO $\mathcal{D}_L$

Proposition 5 (Sinkhorn-Knopp Projection). *Let $\mathbf{X} \in \mathbb{R}_{\geq 0}^{M \times N}$ with $\|\mathbf{X}\|_1 > 0$. The iteration*

1. **Scale:** $\mathbf{X} \leftarrow \mathbf{X} \cdot (L/\|\mathbf{X}\|_1)$.
2. **Row cap:** if $(\mathbf{X}\mathbf{1}_N)_i > 1$, divide row i by $(\mathbf{X}\mathbf{1}_N)_i$.
3. **Col cap:** if $(\mathbf{X}^T\mathbf{1}_M)_j > 1$, divide col j by $(\mathbf{X}^T\mathbf{1}_M)_j$.
4. Repeat 1–3 until convergence.

converges to a feasible point in $\mathcal{D}_L$ when $L \leq \min(M, N)$.

Proof sketch. Steps 2–3 are alternating projections onto $\mathcal{R} = \{\mathbf{X} \geq 0 : \mathbf{X}\mathbf{1} \leq \mathbf{1}\}$ and $\mathcal{C} = \{\mathbf{X} \geq 0 : \mathbf{X}^T\mathbf{1} \leq \mathbf{1}\}$, both convex and closed. Step 1 enforces $\|\mathbf{X}\|_1 = L$. By Dykstra’s theorem, this converges when the intersection is non-empty. The uniform matrix $\frac{L}{MN}\mathbf{1}_{M \times N}$ lies in all three constraint sets, so the intersection is non-empty. $\square$

IX. FULL ALGORITHM

Algorithm 1 WCS-GNCCP with Frank-Wolfe

Require: $\mathbf{A}_G, \mathbf{A}_H$; cost $\mathbf{C}$; size L ; weights α, γ

Ensure: Similarity score $k \in (0, 1]$

- 1: $\hat{\mathbf{A}}_G \leftarrow \mathbf{A}_G / \|\mathbf{A}_G\|_F$; $\hat{\mathbf{A}}_H \leftarrow \mathbf{A}_H / \|\mathbf{A}_H\|_F$ {Prop. 4}
 - 2: $\hat{\mathbf{C}} \leftarrow (\mathbf{C} - \mathbf{C}_{\min}) / (\mathbf{C}_{\max} - \mathbf{C}_{\min})$
 - 3: $\mathbf{X} \leftarrow \frac{L}{MN}\mathbf{1}_{M \times N}$ {Prop. 1(1)}
 - 4: **for** $\zeta = 1, 1 - \Delta\zeta, \dots, -1$ **do**
 - 5: **for** $t = 1, \dots, T_{\text{FW}}$ **do**
 - 6: $\nabla \leftarrow \nabla_{\mathbf{X}} J_{\zeta}$ {Prop. 2}
 - 7: $\mathbf{Y} \leftarrow \text{Hungarian}(\nabla)$ {Def. 4}
 - 8: $g \leftarrow \langle \nabla, \mathbf{X} - \mathbf{Y} \rangle$ {Thm. 2}
 - 9: **if** $g < \varepsilon$ **then**
 - 10: **break**
 - 11: **end if**
 - 12: $\lambda \leftarrow \text{ArmijoLS}(\mathbf{X}, \mathbf{Y}, J_{\zeta})$
 - 13: $\mathbf{X} \leftarrow \text{Sinkhorn}(\mathbf{X} + \lambda(\mathbf{Y} - \mathbf{X}), L)$ {Prop. 5}
 - 14: **end for**
 - 15: **end for**
 - 16: $\mathbf{X}^* \leftarrow \text{Hungarian}(-\mathbf{X})$ {discretize}
 - 17: $F^* \leftarrow F(\mathbf{X}^*)$ {Def. 2}
 - 18: **return** $k = \exp(-\gamma F^* / L)$ {Prop. 3}
-